

Cloud Computing

Lab-7

Name: Monisha Sharma

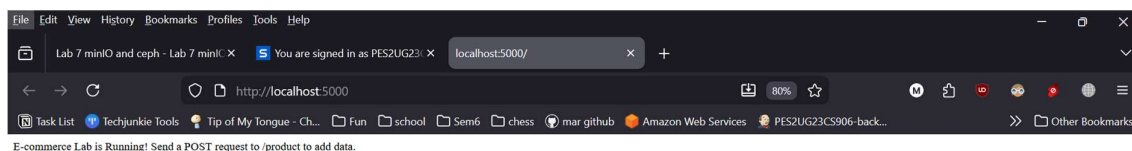
SRN: PES2UG23CS906

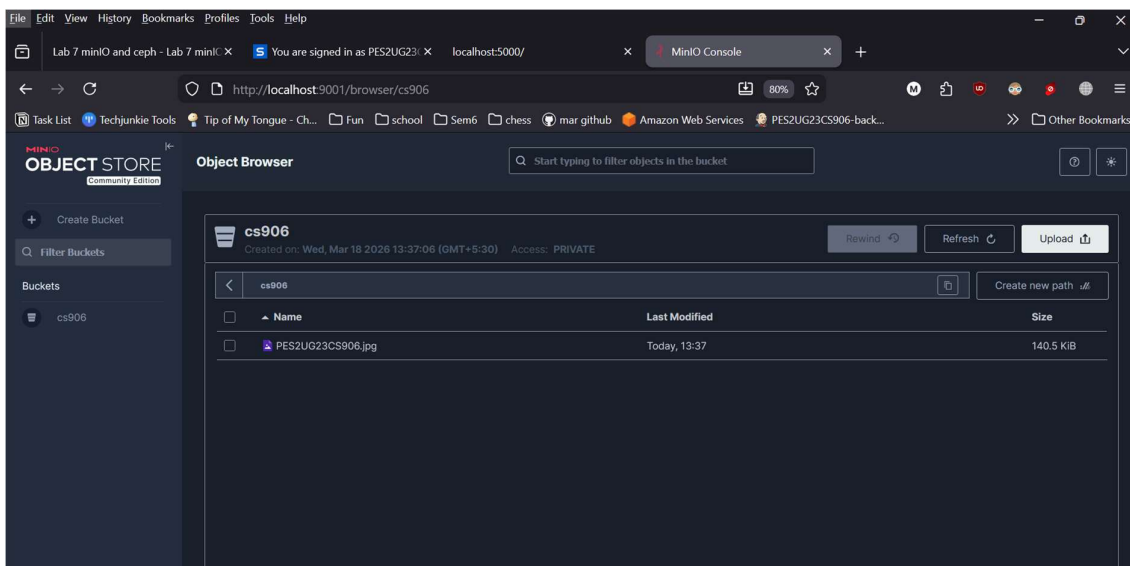
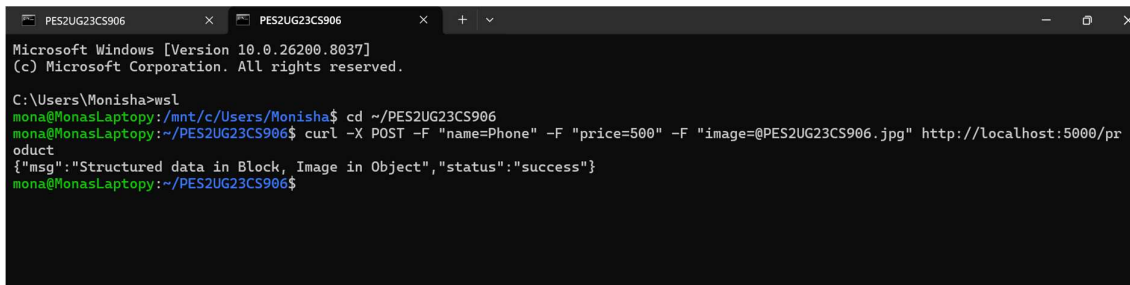
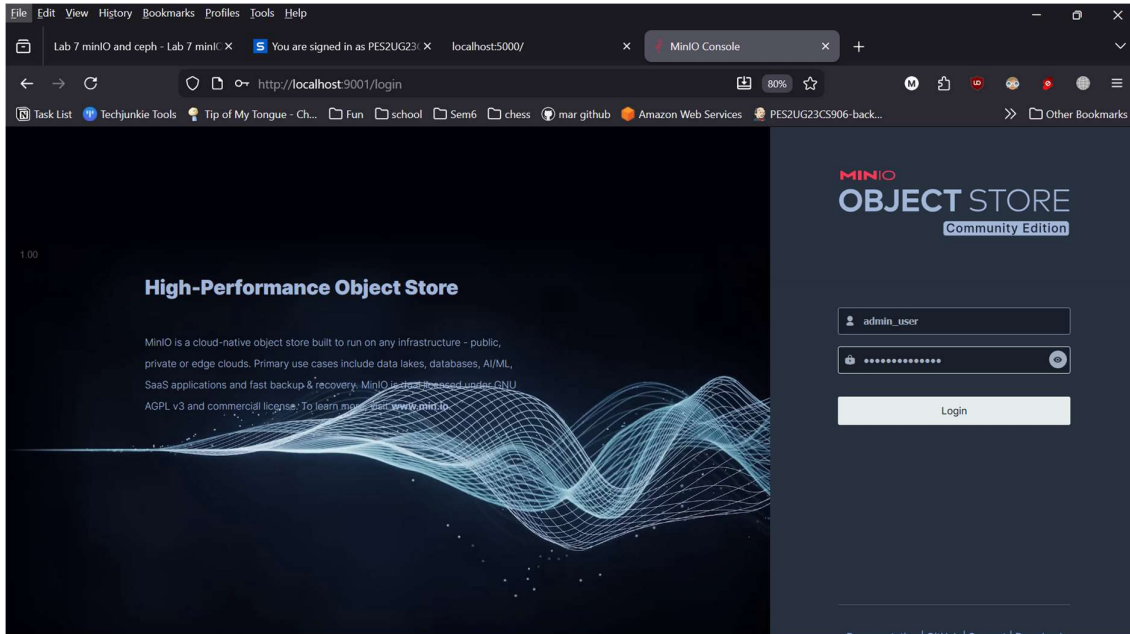
Class: 6F

Date: 18/03/2026

```
PES2UG23CS906
✓ Image minio/minio          Pulled          9.5s
✓ 34013573f278              Pull complete 0.1s
✓ f9c0805c25ee              Pull complete 5.8s
✓ 1008deaf6ec4              Pull complete 6.1s
✓ 0288b5a0d7e7              Pull complete 0.1s
✓ 71e9fc939447              Pull complete 6.2s
✓ c1bc68842c41              Pull complete 0.1s
✓ Image lab-7-storage-webapp Built          30.1s
✓ Network lab-7-storage_default Created         0.1s
✓ Container lab-7-storage-minio-1 Created          0.4s
✓ Container lab-7-storage-webapp-1 Created          0.2s
Attaching to minio-1, webapp-1
minio-1 | INFO: Formatting 1st pool, 1 set(s), 1 drives per set.
minio-1 | INFO: WARNING: Host local has more than 0 drives of set. A host failure will result in data becoming unavailable.
minio-1 | MinIO Object Storage Server
minio-1 | Copyright: 2015-2026 MinIO, Inc.
minio-1 | License: GNU AGPLV3 - https://www.gnu.org/licenses/agpl-3.0.html
minio-1 | Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 linux/amd64)
minio-1 |
minio-1 | API: http://172.19.0.2:9000 http://127.0.0.1:9000
minio-1 | WebUI: http://172.19.0.2:9001 http://127.0.0.1:9001
minio-1 |
minio-1 | Docs: https://docs.min.io
webapp-1 | Starting Flask app on http://0.0.0.0:5000
webapp-1 | * Serving Flask app 'app'
webapp-1 | * Debug mode: off
webapp-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
webapp-1 | * Running on all addresses (0.0.0.0)
webapp-1 | * Running on http://127.0.0.1:5000
webapp-1 | * Running on http://172.19.0.3:5000
webapp-1 | Press CTRL+C to quit
webapp-1 | 172.19.0.1 - - [18/Mar/2026 07:55:50] "GET / HTTP/1.1" 200 -
webapp-1 | 172.19.0.1 - - [18/Mar/2026 07:55:50] "GET /favicon.ico HTTP/1.1" 404 -

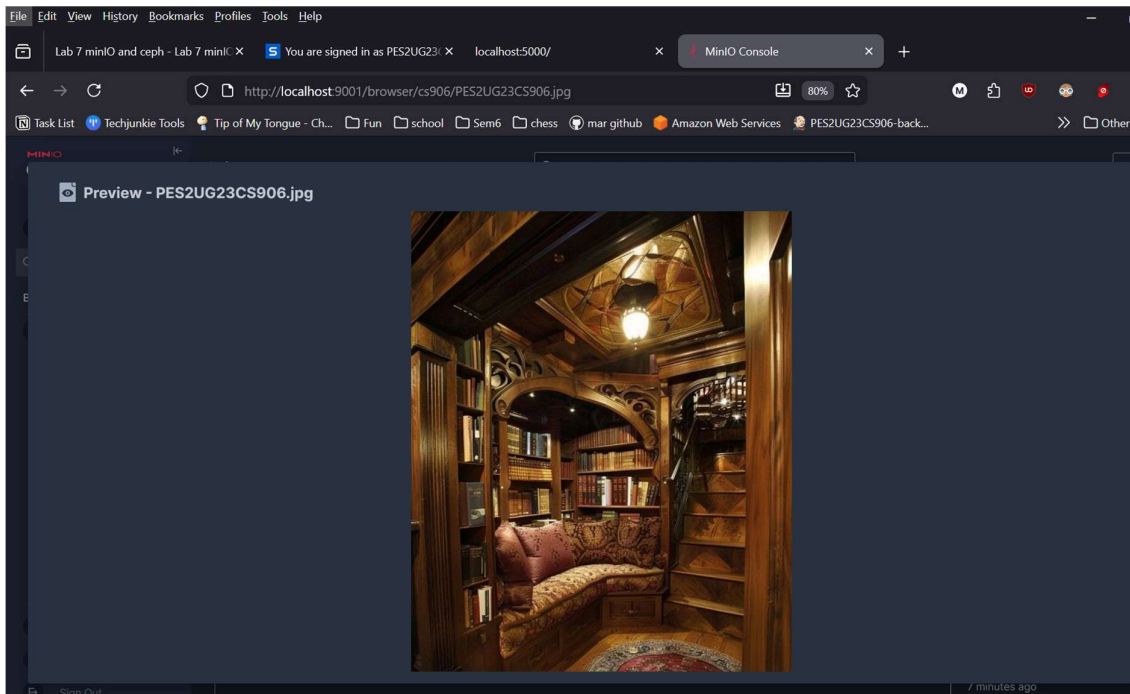
View in Docker Desktop View Config Enable Watch Detach
```





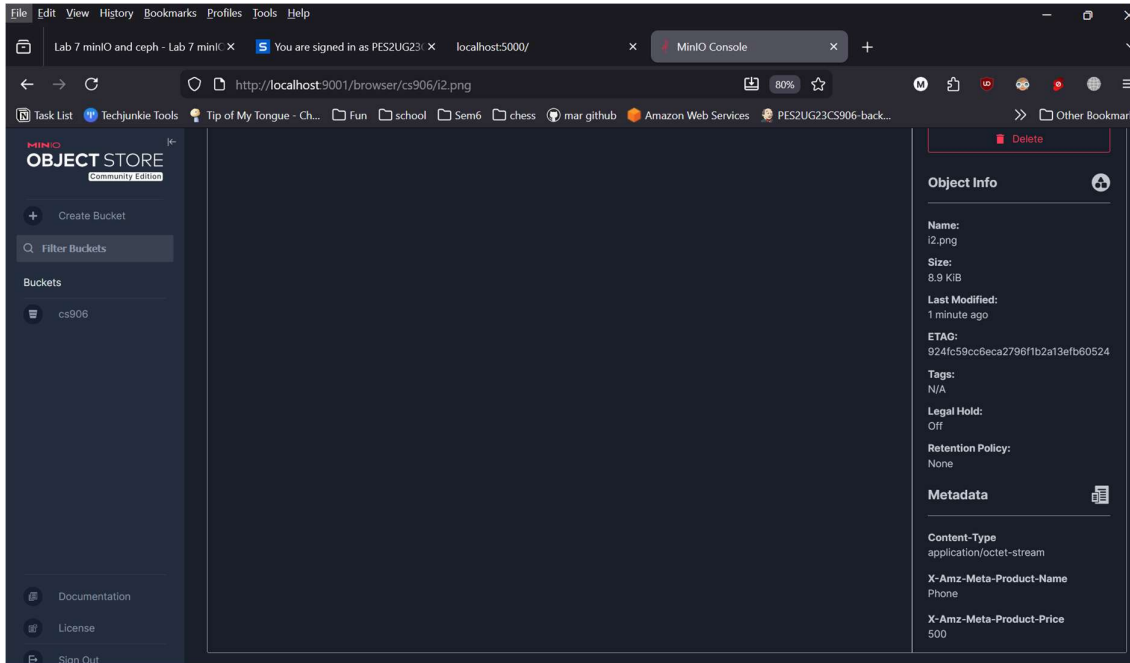
```
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage/my_block_data$ ls
ecommerce.db
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage/my_block_data$
```

```
Error response from daemon: No such container: lab-7-storage-webapp-1
mona@MonasLaptopy:~/PES2UG23CS906$ docker exec -it lab-7-storage-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.close()"
[[1, 'Phone', 500.0, 'PES2UG23CS906.jpg']]
mona@MonasLaptopy:~/PES2UG23CS906$
```



Exp2

```
mona@MonasLaptopy:~/PES2UG23CS906$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@i2.png" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
mona@MonasLaptopy:~/PES2UG23CS906$ docker exec -it lab-7-storage-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.close()"
[[1, 'Phone', 500.0, 'PES2UG23CS906.jpg'], (2, 'Phone', 500.0, 'i2.png')]
mona@MonasLaptopy:~/PES2UG23CS906$
```



Exp3

```
mona@MonasLaptopy:~/PES2UG23CS906$ docker exec -it lab-7-storage-webapp-1 sh -c "dd if=/dev/zero of=/mnt/block_volume/filler.bin bs=1M"
dd: error writing '/mnt/block_volume/filler.bin': No space left on device
1+0 records in
0+0 records out
1040384 bytes (1.0 MB, 1016 KiB) copied, 0.00169631 s, 613 MB/s
mona@MonasLaptopy:~/PES2UG23CS906$ docker exec -it lab-7-storage-webapp-1 df -h /mnt/block_volume
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.0M  1.0M   0 100% /mnt/block_volume
mona@MonasLaptopy:~/PES2UG23CS906$ curl -X POST -F "name=Phone3" -F "price=700" -F "image=@i3.png" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the application.</p>
mona@MonasLaptopy:~/PES2UG23CS906$ |
```

```
PES2UG23CS906 x PES2UG23CS906 x +
=> => exporting manifest sha256:49e193904ba78ff1d28b82df6ff3cf4289ab3f9157c2bc8a6e979c8dcc328ab3 0.0s
=> => exporting config sha256:7222237505545fb4909cae9d63bd1a92b0f1530da6f63a4a8910e101ea74447f 0.0s
=> => exporting attestation manifest sha256:82e87b6463f27cd23cf2d01d7ab83d0389ef7348d16f04c4d260b0b4244b89 0.0s
=> => exporting manifest list sha256:cda52e7075ed362125373ff4dfcfd1b4c0b5566a2770bfe14e1e8fe631a38c8 0.0s
=> => naming to docker.io/library/lab-7-storage-webapp:latest 0.0s
=> => unpacking to docker.io/library/lab-7-storage-webapp:latest 0.0s
=> => resolving provenance for metadata file 0.0s
[+] up 3/3
Image lab-7-storage-webapp Built 12.1s
Container lab-7-storage-minio-1 Running 0.0s
Container lab-7-storage-webapp-1 Recreated 2.0s
Attaching to minio-1, webapp-1
webapp-1 | Starting Flask app on http://0.0.0.0:5000
webapp-1 | * Serving Flask app 'app'
webapp-1 | * Debug mode: off
webapp-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
webapp-1 | * Running on all addresses (0.0.0.0)
webapp-1 | * Running on http://127.0.0.1:5000
webapp-1 | * Running on http://172.19.0.3:5000
webapp-1 | Press CTRL+C to quit
webapp-1 | 172.19.0.1 - - [18/Mar/2026 08:35:50] "POST /product HTTP/1.1" 200 -
webapp-1 | 172.19.0.1 - - [18/Mar/2026 08:40:08] "POST /product HTTP/1.1" 200 -
webapp-1 | [2026-03-18 08:42:09.900] ERROR in app: Exception on /product [POST]
webapp-1 | Traceback (most recent call last):
webapp-1 |   File "/usr/local/lib/python3.9/site-packages/sqlalchemy/engine/base.py", line 1967, in _exec_single_context
webapp-1 |     self.dialect.do_execute(
webapp-1 |   File "/usr/local/lib/python3.9/site-packages/sqlalchemy/engine/default.py", line 952, in do_execute
webapp-1 |     cursor.execute(statement, parameters)
webapp-1 | sqlalchemy.OperationalError: database or disk is full
webapp-1 |
webapp-1 | The above exception was the direct cause of the following exception:
webapp-1 |
webapp-1 | Traceback (most recent call last):
webapp-1 |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 1511, in wsgi_app
webapp-1 |     response = self.full_dispatch_request()
webapp-1 |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 919, in full_dispatch_request
webapp-1 |     rv = self.handle_user_exception(e)
```

Exp4

```
curl: (26) Failed to open/read local data from file/application
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage$ docker-compose stop minio
WARN[0000] /home/mona/PES2UG23CS906/lab-7-storage/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please
remove it to avoid potential confusion
[+] stop 1/1
✓Container lab-7-storage-minio-1 Stopped 0.0s
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage$ curl -X POST -F "name=Phone4" -F "price=800" -F "image=@i3.png" http://localhost:500
0/product
curl: (26) Failed to open/read local data from file/application
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage$
```

```
curl: (26) Failed to open/read local data from file/application
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage$ ls my_block_data
dummy.bin  ecommerce.db
mona@MonasLaptopy:~/PES2UG23CS906/lab-7-storage$
```